

Rababb Pannu

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TECHNICAL SKILLS

Languages: Python, C++, TypeScript, JavaScript, CSS, SQL

Frameworks: PyTorch, TensorFlow, NumPy, pandas, React, Next.js, YOLO, ROS2

Developer Tools: Supabase, Git, Docker, Node.js, Linux, AWS

Hackathon Wins: Hack Canada, Amazon Robotics, Hack The North, UofT RSX SEEK JR, Waterloo Chess Hackathon

EXPERIENCE

Software Developer Intern

Jan. 2026 – April 2026

BMO - NARP Platforms Tech Team

Hybrid: Toronto, ON

- Developing a TSYS Monitoring Dashboard for end-to-end visibility into TSYS data flows, AWS API/MQ volumes, and consumer usage patterns for overall greater observability into all BMO credit card digital infrastructure.
- Developing a File Monitor System/Dashboard using Dynatrace and Ansible to track file presence/metadata, SLA compliance, and send SLA risk notifications, cutting manual escalation time by **60%** and lessening SLA breaches.

Autonomous Software Developer

Sept. 2025 – Present

WATonomous

Waterloo, ON

- Built a real-time robot navigation simulation in Foxglove using ROS2 and C++ for LiDAR-based mapping, containerized with Docker for scalability.
- Improved localization accuracy by **30%** through a refined occupancy grid-based global mapping model for enhanced autonomous navigation.
- Fine tuned a YOLOv11 object detection model for competition specific objects, achieving **95%** accuracy and integrating it into our robotics simulation pipeline.

NeedList.org

Mar. 2026 – Present

Full-Stack Developer (AI Specialist)

Remote: Toronto, ON

- Building end-to-end features for a transparent charitable donation registry (stealth pilot, US & Canada) using TypeScript, NestJS, React, TanStack Query/Router, and Firebase (Auth, Firestore, Cloud Functions, Storage)

PROJECTS

Smart Bin 📷🔗 | Python, TensorFlow, Arduino, Hardware

- Took pictures, labeled, configured training pipeline and evaluated training to finetune a TensorFlow model at StarterHacks 2024 to detect recyclables vs. garbage with **85%** accuracy.
- Integrated the model with Arduino-controlled servo motors using serial communication, enabling automated sorting between waste and recycling in real time.

StudySync 📷🔗 | Python, React, Auth0, MongoDB, Node.js APIs Hack the North 2024 | Best Use of Auth0

- Engineered a full-stack collaborative learning platform enabling users to generate AI-powered notes and quizzes from lip-read videos.
- Achieved **90%** transcription accuracy by integrating Symphonics Labs lip-reading API with OpenAI and Cohere to automatically generate study materials.
- Implemented user authentication and sharing features using Node.js, Auth0, and MongoDB, winning Best Use of Auth0 among **1000+** participants.

FIFA World Cup Graphs Site 📷 | Python, SQL, Matplotlib, pandas, React

- Built an interactive data visualization platform for FIFA World Cup statistics using React with dynamic filtering and real-time graph updates.
- Engineered a backend pipeline using pandas and Matplotlib to query data and generate plots, reducing manual processing time by **95%**.
- Explored SQL queries in MySQL for match statistics, then streamlined the project by shifting to CSV files.

EDUCATION

University of Waterloo

Waterloo, ON

Bachelor of Applied Science in Computer Engineering

Sept. 2025 – Present